

Muhammad Jawad Ali

Software Engineer | AI, Machine Learning & Computer Vision

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Computer Science graduate with hands-on experience in Python, Machine Learning, Deep Learning, Computer Vision, and full-stack AI application development. Built a multi-tenant SaaS AI agent platform and computer vision systems using FastAPI, Next.js, PostgreSQL, RAG, YOLO, OpenCV, TensorFlow, and PyTorch. Experienced in data science workflows, API integrations, Docker-based deployment, and model development. Seeking entry-level opportunities in Software Engineering, AI/ML, Computer Vision, or Data Science.

EDUCATION

BSCS — University of Management and Technology (UMT), Lahore | 2026 | CGPA: 3.00

I.Cs. (Computer Science) — Punjab Group of Colleges (PGC), Lahore | 2019 | 660/1100 (60.90%)

Matriculation — Government Muslim High School, Lahore | 2017 | 880/1100 (80.0%)

Relevant Coursework: Databases, Programming, Algorithms, Operating Systems, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision.

EXPERIENCE

NexSkill, Arfa Karim Tower, Lahore | Data Science Intern | May 2024 – October 2024

- Designed and implemented machine learning workflows involving data cleaning, feature engineering, model development, and performance evaluation using Scikit-learn.
- Developed a customer churn analysis mini-project using Logistic Regression and Decision Tree classifiers.

NexSkill, Arfa Karim Tower, Lahore | Front End Developer | March 2023 – November 2023

- Developed front-end web interfaces using HTML, CSS, and JavaScript.
- Built a customized Spotify-like interface with interactive UI components and dynamic 3D lighting effects.

PROJECTS

Aegis — Multi-Tenant AI Agent Platform (SaaS) | Project Owner

- Led development of a multi-tenant SaaS platform for AI-powered customer support across Web, Facebook, Instagram, WhatsApp, and Email.
- Built backend and AI-powered functionality using FastAPI, PostgreSQL, Next.js, and RAG-based knowledge retrieval.
- Managed Meta API integrations and multi-tenant data security architecture.
- Deployed the production system on a Linux VPS using Docker and Nginx.

Tech Stack: FastAPI, PostgreSQL, Next.js, RAG, Meta APIs, Docker, Nginx, Linux

AI-Powered Decision Review System (DRS) for Local Cricket | Final Year Project — In Progress | Sept 2025 – Present

- Built an AI-based cricket Decision Review System for local and grassroots cricket.
- Implemented ball detection and tracking, trajectory prediction, and LBW/edge decision support from broadcast footage.
- Achieved 94% validation accuracy and designed the system as a low-cost alternative to professional cricket DRS technology.
- [GitHub Repository](#)

Tech Stack: Python, YOLO11, ByteTrack, Kalman Filtering, OpenCV, PyTorch, Jupyter

Automated Car Number Plate Detection | January 2025

- Built an automated vehicle license plate detection and recognition system for images and video feeds.
- Implemented deep learning-based object detection and OCR to identify and extract vehicle plate numbers.

Tech Stack: Python, OpenCV, YOLOv8, EasyOCR

Human Facial Emotion Detection | March 2025

- Built a real-time facial emotion recognition system using a Convolutional Neural Network (CNN).
- Classified facial expressions into seven emotion categories through live webcam input with visual feedback.
- [GitHub Repository](#)

Tech Stack: Python, OpenCV, Keras, TensorFlow, CNN

Intelligent Event Reminder Bot | March 2025

- Developed a smart reminder system that stores event details in Firebase Firestore.
- Implemented rule-driven logic for time-based event triggers and notifications.
- [GitHub Repository](#)

Tech Stack: Python, Firebase Firestore, Firebase Admin SDK, Datetime

TECHNICAL SKILLS

Programming: Python, C, C++, JavaScript

Machine Learning: Scikit-learn, Classification, Prediction, Feature Engineering, Model Evaluation

Deep Learning: TensorFlow, Keras, PyTorch, CNN

Computer Vision: OpenCV, YOLO, Object Detection, Facial Recognition, ByteTrack, Kalman Filtering

Web Development: HTML, CSS, JavaScript, Next.js, FastAPI

Databases: MySQL, SQL Server, PostgreSQL, Firebase Firestore

Mobile Development: Kotlin, XML

DevOps / Tools: Git, Bitbucket, Docker, Nginx, Linux, Jira, Jupyter

Computer Networks: Cisco Packet Tracer, VLANs, DHCP, Router Subinterfaces, Port Security, Wireless Access, Dynamic Routing

HOBBIES & INTERESTS

Cricket, Football, Snooker, Video Games

REFERENCES

Engr. Bashir Ahmad

Senior Software Engineer, Aarna Capital, Dubai, UAE

Email: bashmastr@gmail.com

Professional Reference / Mentor